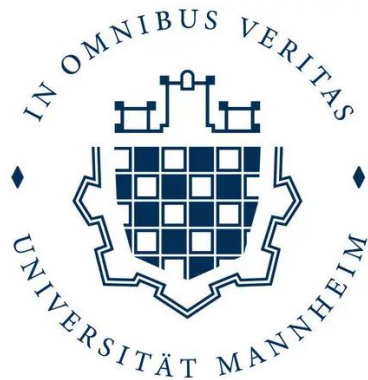




Harnessing GPT for Enhanced Academic Writing: Evidence from a Field Experiment with Early-Career Researchers in the Social Sciences

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Generative AI like GPT is reshaping how researchers write, review, and communicate scholarly work today.

With rapid advancements in AI, tools like GPT enhance writing clarity and coherence, offering new opportunities and challenges for early-career researchers and academia at large. Understanding their impact is critical to harness benefits responsibly while maintaining academic rigor.

Large Language Models and the Academy



- We know AI can improve performance in many general settings.
But in what sense does it help?
- Our focus:
Experts creating expert knowledge.
- A unique collaboration with Springer Nature
Coordination, media outreach, and formed a special issue in Humanities and Social Science Communications for participants.

Context

Why Study GPT in Social Science Writing?



- Social sciences involve complex, nuanced arguments requiring clear and logical expression.
- Intensive, passionate, non-routine research
- Often-routine structures and flow to academic papers
- **Prior evidence**
 - GPT can boost productivity, but how, where, and why?
 - Understanding GPT's impact helps responsibly integrate AI tools in academic workflows, supporting early-career researchers and advancing scientific communication.

Findings: Where are the Opportunities for Academia?



Enhance Clarity and Coherence

Participants who used AI saw an increase in clarity and coherence, as judged by human evaluators



Informative Reviewing and Assessment

AI assessments were informative, though not perfect, as a peer review tool



Maintain Originality

Participants who used AI saw no drop in originality, as judged by human evaluators



Shifting Writing Reflections

Participants who used AI were more likely to characterize their writing using instrumental words (make/find) and less likely to use temporal words (when/now/then).

Challenges

Challenges of AI Posed to the Academy: Is AI Generated Research Any Good?



Reducing Originality



Writing is Thinking



More, but Worse Work



Losing the Human Element



Challenges

Challenges of AI Posed to the Academy: Generate *and* Assess with AI?



Reducing Originality



Writing is Thinking



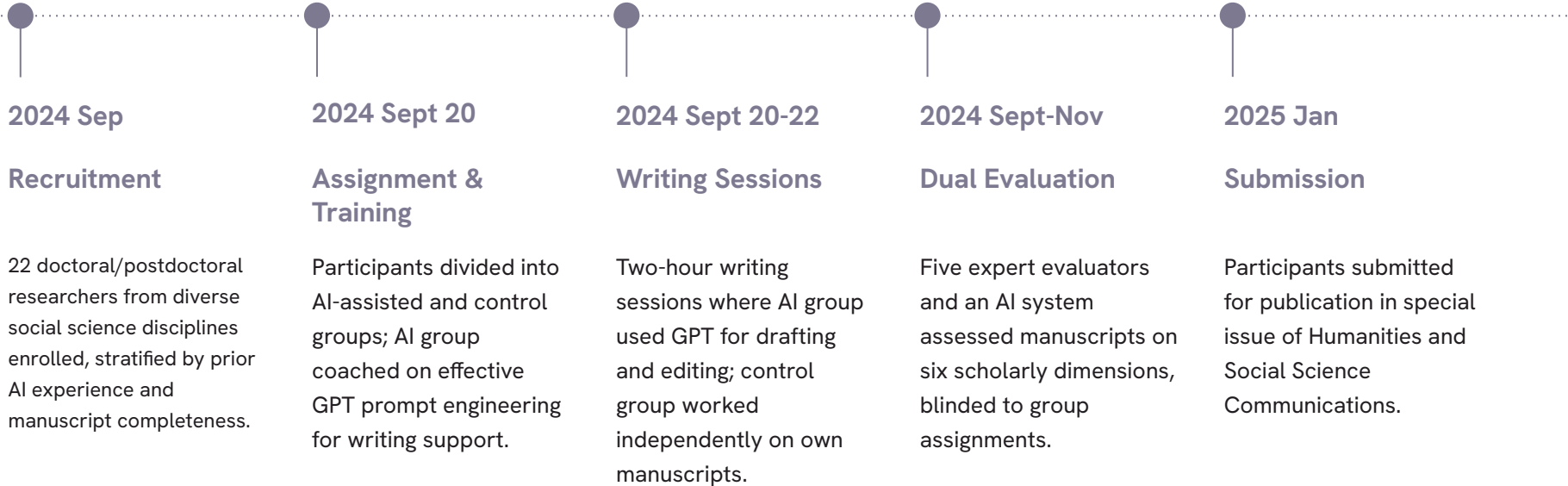
More, but Worse Work



Losing the Human Element



Study Overview: Mannheim Field Experiment



Key Evaluation Dimensions



Clarity

Measures how clearly ideas and arguments are expressed, ensuring readers can easily understand the scientific message.



Analytical Depth

Focuses on the sophistication and thoroughness of analysis and argumentation within the paper.



Coherence

Assesses logical flow and connectivity of arguments, making the paper easy to follow and structurally sound.



Literature Relevance

Judges how well the paper integrates and builds upon existing research and references.



Originality

Evaluates the novelty and uniqueness of ideas, reflecting innovative contributions to the field.



Methodological Rigor

Examines the soundness and precision of research methods and procedures used in the study.

Human vs. AI Assessment



Human Expert Evaluations

- Five faculty experts assessed papers on clarity, coherence, originality, analysis, methodology, literature relevance.
- Used a 5-point Likert scale, blinded to group assignments, ensuring unbiased judgments.
- Captured nuanced, diverse academic perspectives and deep intellectual rigor.
- Differentiated between surface clarity and deeper analytical quality.



AI-Based Evaluations

- AI tool SakanaAI assessed papers using the same criteria and scale as humans.
- AI showed consistency in scoring but favored surface-level clarity improvements.
- Tended to underestimate deeper aspects like originality and methodological rigor.
- Useful for initial screening but lacks sensitivity to complex academic nuances.

Treatment Effect Estimates

Impact of GPT Assistance on Scholarly Dimensions

	Originality	Clarity	Coherence	Depth of Analysis	Literature Review	Methodological Rigor
Estimate	0.11	0.49	0.43	0.07	0.28	0.27
t-statistic	0.61	2.61	2.29	0.37	1.49	1.46
adjusted p	0.87	0.01	0.04	0.94	0.37	0.38

1. Significant Clarity and Coherence gains highlight GPT's selective enhancement of structural writing features.
2. No measurable impact on Originality, Depth of Analysis, Literature Review, or Methodological Rigor

Dimensions of Assessment

Recovering underlying dimensions

	Originality	Clarity	Coherence	Depth of Analysis	Literature Review	Methodological Rigor	Percent Explained
Dimension 1	-0.25	-0.33	-0.41	0.71	-0.09	0.37	0.56
Dimension 2	-0.83	0.47	0.29	0.1	-0.02	-0.02	0.27
Dimension 3	-0.19	-0.13	-0.27	-0.03	0.89	-0.27	0.10

Principal components analysis of human assessors' evaluations revealed three distinct dimensions:

1. Depth and Rigor
2. Clarity and Coherence
3. Literature Review

Human vs. AI: Judgment Gaps



Strengths of AI Evaluation

- AI provides rapid, consistent preliminary assessments of clarity and coherence.
- AI helps streamline initial manuscript screening, saving valuable human time.
- Moderate correlation with human expert ratings supports AI's utility as a complementary tool.



Limitations and Challenges

- AI tends to overlook subtle intellectual depth and methodological rigor.
- Human evaluators reward improvements in clarity and coherence more than AI does.
- AI scores are more homogeneous, lacking human evaluators' nuanced judgment variability.
- Risk of over-reliance on AI may diminish critical human oversight and detailed peer review.

How GPT Changes Reflection



Increased Instrumental Language

Writers using GPT showed a significant rise in practical, task-oriented words like 'make,' 'work,' and 'find', highlighting a focus on action and problem-solving.



Reduced Temporal References

There was a notable decrease in the use of temporal markers such as 'when,' 'now,' and 'then', suggesting less emphasis on narrative flow and more on immediate objectives during writing.

Maintain the Human Element



Accentuate, Don't Eliminate

The emphasis was on using AI to strengthen and sharpen your argument, as an iterative tool



Recursive Editing

Writing with GPT where the researcher and AI recursively switch roles as Creator and Editor



Delegate Coding

It's *really* good at this.

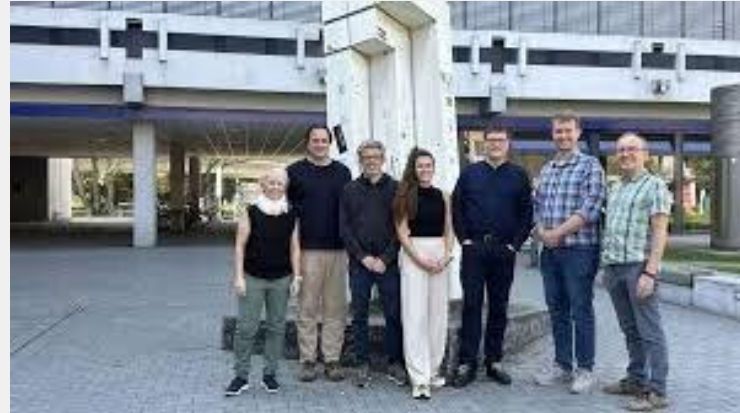
Implications

ACCENTUATE the Human Element



Have fun!

The event worked with only 22 people because we had fun: regular breaks for sport, good food and coffee, laughter, regular trips around Mannheim.



AI can enhance structure, but human insight ensures originality and rigor.

While GPT improves clarity and coherence in academic writing, it cannot replace the nuanced judgment, creativity, and critical thinking that only human experts provide. Responsible integration of AI requires preserving the human role in scholarly evaluation and interpretation.

